

- Q.22 Write the various function of transmission system.
 Q.23 What are the requirements of clutch?
 Q.24 Write a short note on differential.
 Q.25 Write the advantages & disadvantages of light alloy cast wheel.
 Q.26 Explain the different types of wheels
 Q.27 Write down the function of suspension system.
 Q.28 Write a short note on full-floating axle.
 Q.29 Explain power steering in brief.
 Q.30 Explain the working of Hydraulic brake system
 Q.31 Write functions of battery in an automobile.
 Q.32 Give the comparison of dynamo and alternator.
 Q.33 Write a short note on magneto ignition system.
 Q.34 Explain high rate discharge test of battery.
 Q.35 What is universal joint? Explain its function.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain construction & working of synchromesh gear box with sketch.
 Q.37 Explain the construction, working and advantage of vacuum brakes.
 Q.38 What is an alternator? Give its principle, construction & working with neat sketch.

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5th Sem / Mech. Engg. (MSIL) Sub : Automobile Engineering

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Function of CAM shaft is _____
 a) To operate the carburetor
 b) To operate fuel injector
 c) To operate inlet & exhaust valves
 d) None of these
- Q.2 Automobile can be classified based on which of the following parameter?
 a) Fuel used b) Transmission
 c) Drive d) All of these
- Q.3 Which of the following is not a part of the transmission system?
 a) Clutch b) Wheels
 c) Gear Box d) Axles
- Q.4 What is the function of the alternator?
 a) Recharging the battery
 b) Voltage regulator
 c) Auto ignition
 d) None of these

- Q.5 Which of the following is found in an automobile's electrical system?
- Lighting system
 - Battery
 - Alternators
 - All of these
- Q.6 The temperature of the piston will be more at _____ in an automobile engine.
- The piston rings
 - The piston wall
 - The crown of the piston
 - The skirt of the piston
- Q.7 If there are seven clutch plates in a multi-plate clutch, what are the numbers of pairs of contact surfaces?
- 5
 - 4
 - 6
 - 8
- Q.8 In which type of manual transmission the double de-clutching is used?
- Constant mesh gear box
 - Sliding mesh gear box
 - Synchromesh gear box
 - Epicyclical gear box
- Q.9 What is the purpose of supercharger?
- Increase the pressure of inlet charge
 - Increase the speed of engine
 - Increase gear ratio
 - Increase the cooling rate

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- Q.10 The function of ABS is that
- Reduce the stopping distance
 - Minimizes the brake fade
 - Maintain directional control during braking by preventing the wheels from locking
 - Prevents nose dives during braking and thereby postpones locking of the wheels

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name the components of an automobile.
- Q.12 What is rear wheel drive?
- Q.13 What is the purpose of over drive?
- Q.14 What is automatic transmission?
- Q.15 What is the value of king pin inclination?
- Q.16 Name the different parts of wire wheel.
- Q.17 Write the function of steering system.
- Q.18 Parking brakes are meant to hold the vehicle on a slope. (True/False)
- Q.19 Sprung weight = weight of body & chassis + ____.
- Q.20 Write the full form of CC.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a short note on hybrid electric vehicles.

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